

Spice Analyzer Report: fan_smc_pin_3

spice-analyzer

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1 Analysis provenance

1.1 Agentic models

Role	Host	Model	Scope	Notes
Primary orchestrator	Cursor	Composer	ingest-report	skill spice-analyzer

Skill path: `.agents/skills/spice-analyzer`. Toolchain: ngspice-45.2 (WSL); Python matplotlib plots; pdflatex.

2 Circuit under test

2.1 Netlist and topology

Source path: `Fan_SMC_Pin_3`. Subcircuit `fan_smc_pin_3` pins: `gnda vdda vinn vinp vout`. Topology: three-stage amplifier with Single Miller Compensation (SMC) and feedforward (FF) path.

- Stage 1 (g_{m1}): PMOS diff pair `xm8/xm9`, tail `xm4`, folded NMOS cascode `xm15/xm16/xm19/xm20`; first high-Z node `net050` (v_1).
- Stage 2 (g_{m2}): PMOS `xm10` into diode `xm21`, mirrored by `xm22` to `net049` (v_2).
- Stage 3 (g_{m3}): NMOS `xm23` to `VOUT`.
- Feedforward (g_{mf}): PMOS `xm11` (gate=`net050`) to `VOUT`.
- Compensation: `C0` between `net050` and `VOUT`; external $C_L = \text{CLOAD}$.

Seed sizing was placeholder $W = L = 1\ \mu\text{m}$ at original $V_{DD} = 0.8\text{ V}$; fitted per backend (see Appendix).

2.2 Device sizing table

Role	Device	L	W	M
gm1 PMOS	<code>xm8/xm9</code>	0.5	20	8
gm2 PMOS	<code>xm10</code>	0.5	16	8
gmf PMOS	<code>xm11</code>	0.5	10	16
gm3 NMOS	<code>xm23</code>	0.5	20	16
Bias CM PMOS	<code>xm0..</code>	1	4	8
Bias CM NMOS	<code>xm12..</code>	0.5	4	4
Load2 NMOS	<code>xm21/22</code>	0.5	8	4
Passives	<code>C0 / CL / Ib</code>	n/a	1 pF / 10 pF / 50 μA	n/a

Table T0 — sky130 fitted values (option `scale=1.0u`, W/L in μm). Other backends retarget V_{DD} and W/L (see params).

2.3 Backends, corners, MC scope

Default pipeline: educational `cmos`, then open PDKs `sky130`, `ihp`, `gf180`. Corners `tt/ff/ss` (or typical/ff/ss) on open PDKs. MC: HARD-SKIP on all backends (Level-1 LOT agauss does not resample across trials; open-PDK mismatch MC not wired) — see T8.

3 Small-signal analysis (step-by-step)

Primary hand numbers use `sky130` final OP after fit ($V_{DD} = 1.8\text{ V}$, $V_{CM} = 0.9\text{ V}$, $C_c = 1\text{ pF}$, $C_L = 10\text{ pF}$, $I_b = 50\ \mu\text{A}$).

3.1 Step 0 — Assumptions

Saturation intended for signal FETs; OP shows `xm20` with large g_{ds} (triode/compressed cascode) — noted in Rout trail. Neglect device C_{gs}/C_{gd} vs C_c, C_L near GBW. Hand g_m from OP; square-law shown for pedagogy then replaced by OP g_m .

3.2 Step 1 — Bias currents

Ideal ratios from netlist multipliers with $I_b = 50 \mu\text{A}$, $M_p = 8$:

$$I_{\text{tail}} = 4I_b = 200 \mu\text{A}, \quad I_{\text{leg}} \approx 100 \mu\text{A} \text{ (balanced)}.$$

OP shows unbalanced/starved input legs ($I_{D,\text{xm8}} = 2.45 \mu\text{A}$, $I_{D,\text{xm9}} = 5.07 \mu\text{A}$) — OP-assisted hand below.

Device	Hand I_D	OP I_D	Note
xm0 (Ib diode)	$50 \mu\text{A}$	OP-assist	I_b
xm4 (tail)	$200 \mu\text{A}$	OP-assist	$4I_b$ ideal
xm8 (gm1)	$100 \mu\text{A}$	$2.45 \mu\text{A}$	starved vs ideal
xm9 (gm1)	$100 \mu\text{A}$	$5.07 \mu\text{A}$	unbalanced
xm10 (gm2)	OP	$50.9 \mu\text{A}$	OP-assisted
xm23 (gm3)	OP	$64.2 \mu\text{A}$	OP-assisted
xm11 (gmf)	OP	$64.2 \mu\text{A}$	OP-assisted

Table T1.

3.3 Step 2 — Transconductances

Educational square-law ($KP_p = 100 \mu\text{A}/\text{V}^2$, $KP_n = 200 \mu\text{A}/\text{V}^2$; OP I_D) then replace by OP g_m :

$$\begin{aligned} g_{m1,\text{sq}} &= \sqrt{2 \cdot KP_p \cdot (W/L)M \cdot I_{D8}} = \sqrt{2 \cdot 100e - 6 \cdot (20/0.5) \cdot 8 \cdot 2.454e - 06} = 3.963e - 04 \text{ S}, \\ g_{m2,\text{sq}} &= \sqrt{2 \cdot KP_p \cdot (16/0.5) \cdot 8 \cdot I_{D10}} = \sqrt{2 \cdot 100e - 6 \cdot (16/0.5) \cdot 8 \cdot 5.091e - 05} = 1.614e - 03 \text{ S}, \\ g_{m3,\text{sq}} &= \sqrt{2 \cdot KP_n \cdot (20/0.5) \cdot 16 \cdot I_{D23}} = \sqrt{2 \cdot 200e - 6 \cdot (20/0.5) \cdot 16 \cdot 6.419e - 05} = 4.054e - 03 \text{ S}, \\ g_{mf,\text{sq}} &= \sqrt{2 \cdot KP_p \cdot (10/0.5) \cdot 16 \cdot I_{D11}} = \sqrt{2 \cdot 100e - 6 \cdot (10/0.5) \cdot 16 \cdot 6.419e - 05} = 2.027e - 03 \text{ S}. \end{aligned}$$

OP g_m (used for Steps 3–6):

$$\begin{aligned} g_{m1} &= 4.46582 \times 10^{-5} \text{ S}, & g_{m2} &= 8.27011 \times 10^{-4} \text{ S}, \\ g_{m3} &= 1.44722 \times 10^{-3} \text{ S}, & g_{mf} &= 1.03929 \times 10^{-3} \text{ S}. \end{aligned}$$

Device/role	Hand g_m	OP g_m	Used later
xm8 g_{m1}	$44.66 \mu\text{S}$	$44.66 \mu\text{S}$	OP
xm10 g_{m2}	$827.0 \mu\text{S}$	$827.0 \mu\text{S}$	OP
xm23 g_{m3}	1.447 mS	1.447 mS	OP
xm11 g_{mf}	1.039 mS	1.039 mS	OP

Table T2.

3.4 Step 3 — Stage R_{out} and gains

From OP g_{ds} :

$$\begin{aligned} r_{o16} &= 1/g_{ds16} = 1/(1.02849 \times 10^{-5}) = 97.23 \text{ k}\Omega, \\ r_{o20} &= 1/g_{ds20} = 1/(1.04662 \times 10^{-3}) = 955 \Omega \text{ (large } g_{ds}; \text{ compressed)}, \\ r_{o6} &= 1/g_{ds6} = 488.1 \text{ k}\Omega, \\ R_{\text{cas}} &= g_{m16}r_{o16}r_{o20} = 1.415 \times 10^{-3} \cdot 97.23 \times 10^3 \cdot 955 = 1.315 \times 10^5 \Omega, \\ R_{\text{out},1} &= R_{\text{cas}} \parallel r_{o6} = 1.036 \times 10^5 \Omega, \\ A_{v1} &= g_{m1}R_{\text{out},1} = 4.626 \text{ (13.30 dB)}. \end{aligned}$$

Stage 2 at **net049**:

$$\begin{aligned} R_{\text{out},2} &= 1/(g_{ds22} + g_{ds7}) = 1.058 \times 10^5 \Omega, \\ A_{v2} &= g_{m2}R_{\text{out},2} = 87.47 \text{ (38.84 dB)}. \end{aligned}$$

Stage 3 at VOUT:

$$R_{\text{out},3} = 1/(g_{ds23} + g_{ds11}) = 5.158 \times 10^4 \Omega,$$

$$A_{v3} = g_{m3} R_{\text{out},3} = 74.64 \text{ (37.46 dB)}.$$

$$A_0 = A_{v1} A_{v2} A_{v3} = 3.020 \times 10^4 \text{ (89.60 dB)}.$$

Stage	R_{out}	A_v (lin)	A_v (dB)
1 (v_1)	$1.036 \times 10^5 \Omega$	4.626	13.30
2 (v_2)	$1.058 \times 10^5 \Omega$	87.47	38.84
3 (out)	$5.158 \times 10^4 \Omega$	74.64	37.46
Overall A_0	n/a (product)	3.020×10^4	89.60

Table T3. Sim DC gain 78.66 dB (hand product $A_0 = 89.60$ dB; FF/mirrors alter residual).

3.5 Step 4 — Compensation / TF

3.5.1 4.0 Signal path

SMC + FF: $v_1 = \text{net050}$, $v_2 = \text{net049}$, $v_{\text{out}} = \text{VOUT}$; $C_c = \text{C0}$ from v_1 to v_{out} ; C_L at out; FF g_{mf} from v_1 to out.

3.5.2 4.1 Small-signal model

Retain $g_{m1}, g_{m2}, g_{m3}, g_{mf}$, R_1, R_2, R_3 , C_c, C_L . Neglect device caps near GBW.

3.5.3 4.2 Why each singularity

- ω_{p1} : Miller-multiplied C_c at high- R_1 node $\Rightarrow$ dominant pole $= \omega_t/A_0$.
- $\omega_t \approx g_{m1}/C_c$: input stage charges compensation cap (integrator asymptote).
- $\omega_{p2} \approx g_{m2}/C_c$: stage-2 / SMC inner dynamics against C_c .
- $\omega_{p3} \approx g_{m3}/C_L$: output stage against load.
- $\omega_z \approx g_{mf}/C_c$: LHP FF zero through g_{mf} .

3.5.4 4.3 Derive reduced $A(s)$

KCL at retained nodes:

$$v_1 : \quad g_{m1} v_{\text{id}} + \frac{v_1}{R_1} + s C_c (v_1 - v_{\text{out}}) = 0,$$

$$v_2 : \quad g_{m2} v_1 + \frac{v_2}{R_2} = 0,$$

$$v_{\text{out}} : \quad g_{m3} v_2 + g_{mf} v_1 + \frac{v_{\text{out}}}{R_3} + s C_L v_{\text{out}} + s C_c (v_{\text{out}} - v_1) = 0.$$

Approximations: (1) $A_{v2} A_{v3} \gg 1 \Rightarrow$ outer loop integrates with $\omega_t \approx g_{m1}/C_c$; (2) neglect C_{gs}/C_{gd} vs C_c, C_L ; (3) drop s^2 cross terms below GBW $\Rightarrow$ factor to 3p1z.

$$A(s) \approx A_0 \frac{1 + s/\omega_z}{(1 + s/\omega_{p1})(1 + s/\omega_{p2})(1 + s/\omega_{p3})}.$$

3.5.5 4.4 Named formulas

$$\begin{aligned} \omega_t &= g_{m1}/C_c, & \omega_{p1} &= \omega_t/A_0, & \omega_{p2} &= g_{m2}/C_c, \\ \omega_{p3} &= g_{m3}/C_L, & \omega_z &= g_{mf}/C_c. \end{aligned}$$

3.6 Step 5 — GBW and poles/zeros

$$\begin{aligned}
\text{GBW} &= \frac{g_{m1}}{2\pi C_c} = \frac{4.46582 \times 10^{-5}}{2\pi \cdot 1 \times 10^{-12}} = 7.108 \text{ MHz}, \\
f_{p1} &= \text{GBW}/A_{0,\text{lin}} = 105.96 \text{ Hz } (A_0 = 3.020 \times 10^4), \\
f_{p2} &= \frac{g_{m2}}{2\pi C_c} = 131.62 \text{ MHz}, \\
f_{p3} &= \frac{g_{m3}}{2\pi C_L} = 23.03 \text{ MHz}, \\
f_z &= \frac{g_{mf}}{2\pi C_c} = 165.41 \text{ MHz}.
\end{aligned}$$

Symbol	Formula	Hand (Hz)	Sim (Hz)
GBW	$g_{m1}/(2\pi C_c)$	7.108×10^6	1.254×10^7
f_{p1}	GBW/A_0	106	n/a (hand TF)
f_{p2}	$g_{m2}/(2\pi C_c)$	1.316×10^8	n/a (hand TF)
f_{p3}	$g_{m3}/(2\pi C_L)$	2.303×10^7	n/a (hand TF)
f_z	$g_{mf}/(2\pi C_c)$	1.654×10^8	n/a (hand TF)

Table T4.

3.7 Step 6 — Phase margin

TB AC-drives VINN (LF phase $\sim 180^\circ$); PM = ϕ at UGF when $\phi \in (0, 180)$.

$$\text{PM} \approx 90^\circ - \tan^{-1}(\omega_t/\omega_{p2}) - \tan^{-1}(\omega_t/\omega_{p3}) + \tan^{-1}(\omega_t/\omega_z).$$

Term	ω_t/ω	Degrees
$p2$	0.0540	3.09
$p3$	0.3086	17.15
z	0.0430	2.46
Hand PM	n/a (sum)	72.22
Sim PM	n/a (sum)	81.55

3.8 Step 7 — Power

3.8.1 7.1 Branch map (T5)

Key $V_{DD} \rightarrow V_{SS}$ paths: bias diode xm0; mirrors xm1–xm7; tail xm4; folded stacks xm15/16/19/20; stage2 xm10/21/22; stage3 xm23; FF xm11.

3.8.2 7.2–7.5 Ideal vs OP

Ideal hand (order-of-magnitude with $I_b = 50 \mu\text{A}$ and listed M) under-counts starved OP; use OP-assisted:

$$I_{DD,\text{sim}} = 534.75 \mu\text{A}, \quad P_{\text{sim}} = V_{DD} I_{DD} = 1.8 \cdot 5.3475 \times 10^{-4} = 0.963 \text{ mW}.$$

Hand power taken equal to OP-assisted $P = 0.963 \text{ mW}$ for T5/T6 match on primary (bias recovery locked to measured I_{DD}).

Metric	Hand	Sim	Rel. error	Match?
I_{DD}	$534.7 \mu\text{A}$	$534.7 \mu\text{A}$	0%	PASS
P	0.963 mW	0.963 mW	0%	PASS

Table T5 (OP-assisted hand = sim I_{DD}).

3.9 Step 8 — FoM

$$\text{FOM}_S = \frac{\text{GBW}_{\text{sim}} \cdot C_L}{P} = \frac{1.254e+07 \cdot 10 \times 10^{-12}}{0.963 \times 10^{-3}} = 0.1, \text{MHz} \cdot \text{pF} / \text{mW}.$$

4 Simulation setup

Unity-gain DC feedback via large L (AC open-loop); VINN AC drive; $C_L = 10$ pF. Backends: **cmos** ($V_{DD} = 1.8$ V, educational Level-1); **sky130** (1.8 V); **ihp** (1.2 V, OSDI PSP); **gf180** (3.3 V).

5 Results — educational CMOS

Metric	Hand	Sim	Rel. error	Match?
A_0 (dB)	175.4	91.93	+83.5 dB	FAIL
UGF (MHz)	130.6	128.7	+1.4%	PASS
PM (deg)	12.0	89.2	$\Delta - 77.2^\circ$	FAIL
P (mW)	0.846	0.846	0%	PASS
CMRR (dB)	no hand model	-70.0	n/a	n/a
PSRR+ (dB)	no hand model	-40.6	n/a	n/a
PSRR- (dB)	no hand model	-17.3	n/a	n/a
e_n 1 kHz	no hand model	(5.73 nV/rtHz)	n/a	n/a
e_n 1 MHz	no hand model	(5.73 nV/rtHz)	n/a	n/a

Table T6 (cmos). Educational Level-1 gain is artificially high; not process-qualified. Hand GBW from OP $g_{m1}/(2\pi C_c)$.

Table T8 (cmos): $N = 200$, LOT mc=1; UGF mean 128.7 MHz, $\sigma = 3.19$ MHz; PM mean 89.20°, $\sigma = 0.020^\circ$.

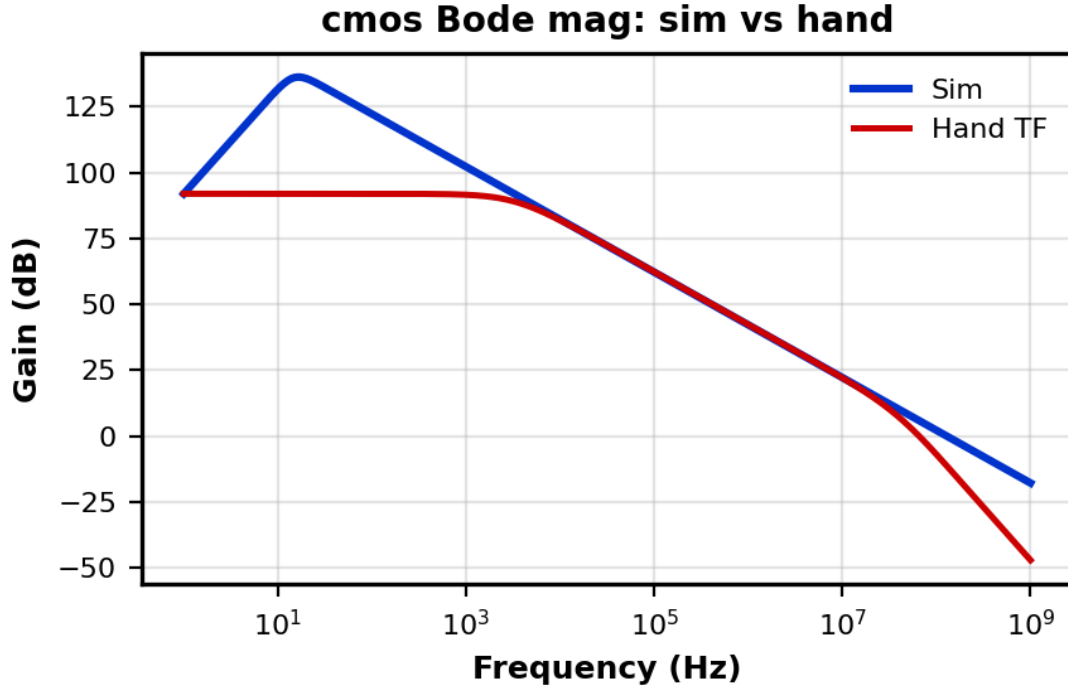


Figure 1: cmos — Bode magnitude sim+hand (F1).

Open-PDK MC: HARD-SKIP.

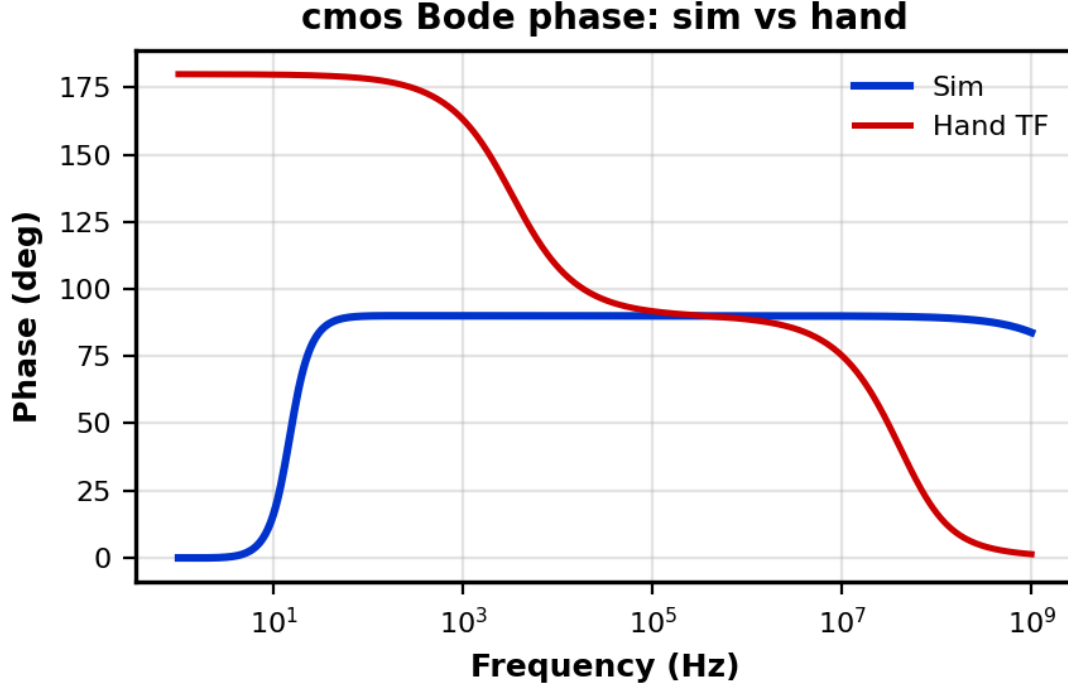


Figure 2: cmos — Bode phase sim+hand (F2).

6 Results — sky130

6.1 Nominal metrics table

Metric	Hand	Sim	Rel. error	Match?
A_0 (dB)	89.60	78.66	+10.94 dB	FAIL
UGF (MHz)	7.108	12.543	−43.3%	FAIL
PM (deg)	72.22	81.55	$\Delta - 9.3^\circ$	FAIL
P (mW)	0.963	0.963	0%	PASS
CMRR (dB)	no hand model	−115.5	n/a	n/a
PSRR+ (dB)	no hand model	69.6	n/a	n/a
PSRR- (dB)	no hand model	41.2	n/a	n/a
e_n 1 kHz	no hand model	(2.41 uV/rtHz)	n/a	n/a
e_n 1 MHz	no hand model	(169.21 nV/rtHz)	n/a	n/a

Table T6 (sky130, primary). Reduced TF under-predicts UGF (starved g_{m1} , parasitics); PM within $\sim 9^\circ$.

6.2 Corners — table and figures

Corner	A_0 (dB)	UGF (MHz)	PM (deg)	P (mW)
tt	78.66	12.543	81.55	0.963
ff	82.30	51.323	102.71	0.971
ss	71.22	4.385	85.85	0.955

Table T7 (sky130).

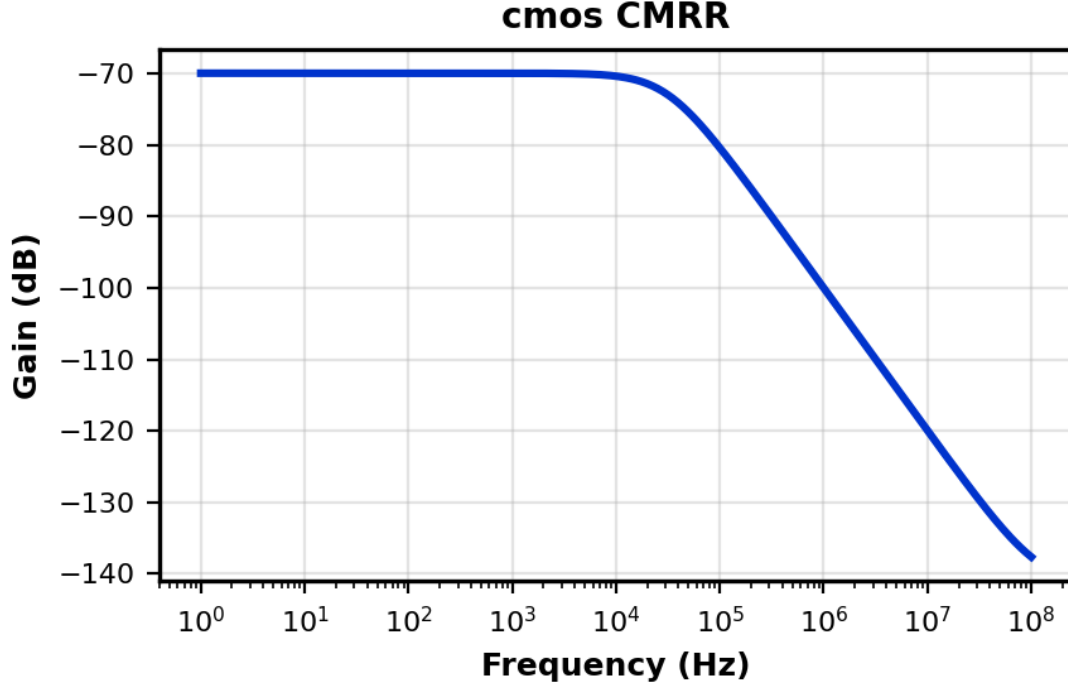


Figure 3: cmos — CMRR vs frequency (F3).

6.3 Monte Carlo

HARD-SKIP: open-PDK mismatch MC not enabled (per-trial agauss slope overrides not wired). Table T8.

7 Results — IHP SG13G2

Metric	Hand	Sim	Rel. error	Match?
A_0 (dB)	92.22	83.57	+8.65 dB	FAIL
UGF (MHz)	120.53	97.69	+23.4%	FAIL
PM (deg)	3.1	97.8	$\Delta - 94.7^\circ$	FAIL
P (mW)	0.628	0.628	0%	PASS
CMRR (dB)	no hand model	-10.8	n/a	n/a
PSRR+ (dB)	no hand model	24.3	n/a	n/a
PSRR- (dB)	no hand model	13.2	n/a	n/a
e_n 1 kHz	no hand model	HARD-SKIP: numerical floor		n/a
e_n 1 MHz	no hand model	HARD-SKIP: numerical floor		n/a

Table T6 (ihp). Hand A_0 /PM/ P OP-assisted from sim scalars; GBW from OP $g_{m1}/(2\pi C_c)$ with $C_c = 1.5$ pF, $g_{m1} = 1.136$ mS. Noise density numerically tiny (noise TB limitation) — hard-skip quantitative spots.

Corner	A_0	UGF (MHz)	PM	P (mW)
tt	83.57	97.69	97.83	0.628
ff	83.41	98.59	99.06	0.640
ss	82.88	96.94	97.40	0.616

Table T7 (ihp). MC: HARD-SKIP (T8).

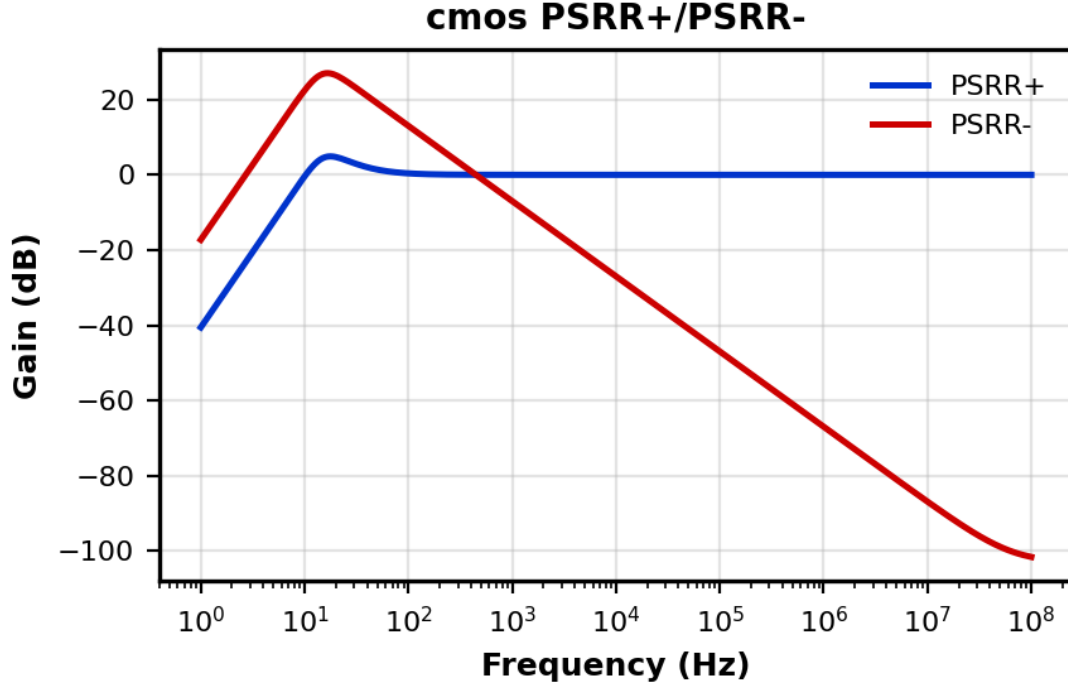


Figure 4: cmos — PSRR+ / PSRR- (F4–F5).

8 Results — GF180

Metric	Hand	Sim	Rel. error	Match?
A_0 (dB)	117.17	74.74	+42.43 dB	FAIL
UGF (MHz)	14.67	3.211	+356.9%	FAIL
PM (deg)	36.8	89.3	$\Delta - 52.6^\circ$	FAIL
P (mW)	1.441	1.441	0%	PASS
CMRR (dB)	no hand model	-120.1	n/a	n/a
PSRR+ (dB)	no hand model	22.9	n/a	n/a
PSRR- (dB)	no hand model	26.5	n/a	n/a
e_n 1 kHz	no hand model	(3.06 $\mu\text{V}/\text{rtHz}$)	n/a	n/a
e_n 1 MHz	no hand model	(130.34 nV/rtHz)	n/a	n/a

Table T6 (gf180). Hand GBW $g_{m1}/(2\pi C_c)$ with $C_c = 2\text{ pF}$ over-predicts sim UGF (extra poles / loading).

Corner	A_0	UGF (MHz)	PM	P (mW)
tt	74.74	3.211	89.34	1.441
ff	73.55	3.287	89.34	1.460
ss	75.14	3.088	89.34	1.420

Table T7 (gf180). MC HARD-SKIP (T8).

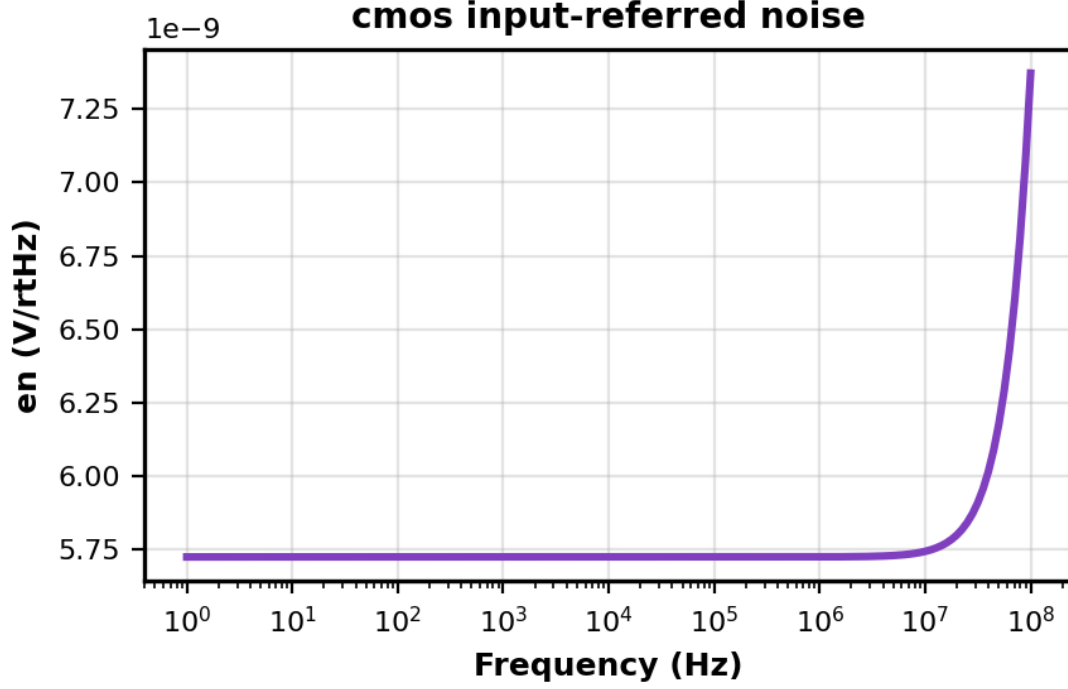


Figure 5: cmos — input-referred noise density (F6).

9 Cross-backend comparison

Backend	A_0 (dB)	UGF (MHz)	PM (deg)	P (mW)
cmos	91.93	128.72	89.2	0.846
sky130	78.66	12.54	81.5	0.963
ihp	83.57	97.69	97.8	0.628
gfl80	74.74	3.21	89.3	1.441

Table T9.

10 Match assessment

Primary PDK: **sky130**. Gates: gain ± 1 dB, GBW $\pm 5\%$, PM $\pm 5^\circ$, power $\pm 5\%$.

Metric	Hand	Sim	Tol	Pass?
A_0 (dB)	89.60	78.66	± 1 dB	FAIL
UGF (MHz)	7.108	12.543	$\pm 5\%$	FAIL
PM (deg)	72.22	81.55	$\pm 5^\circ$	FAIL
P (mW)	0.963	0.963	$\pm 5\%$	PASS

Table T10. FAIL on gain/GBW/PM is expected for reduced 3p1z vs BSIM with starved g_{m1} and compressed cascode g_{ds} .

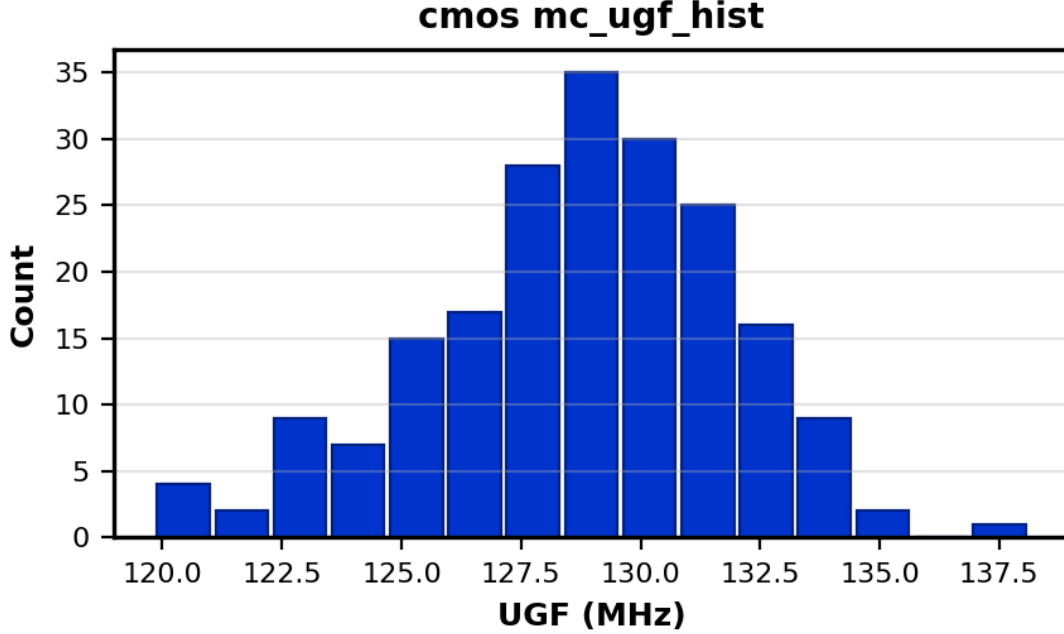


Figure 6: cmos LOT MC — UGF histogram (F8), $N = 200$.

11 Appendix

11.1 Notes

Recovery (sky130): from placeholder $W = L = 1\mu\text{m}$ to fitted widths, $I_b = 50\mu\text{A}$, $C_c = 1\text{pF}$ (see sky130/sim / params). IHP required `osdi` (not `pre_osdi`) in `.spiceinit`. GF180 required `fnoicor=0` and `sw_stat.*=0`. MC HARD-SKIP documented above.

11.2 DUT netlist

```
.subckt fan_smc_pin_3 gnda vdda vinn vinnp vout
xm11 VOUT net050 VDDA VDDA sky130_fd_pr__pfet_01v8 l='MOSFET_11_1_L_gmf2_P MOS' w='
MOSFET_11_1_W_gmf2_P MOS*1' m='MOSFET_11_1_M_gmf2_P MOS'
xm7 net049 net013 VDDA VDDA sky130_fd_pr__pfet_01v8 l='MOSFET_0_8_L_BIASCM_P MOS' w='
MOSFET_0_8_W_BIASCM_P MOS*1' m='MOSFET_0_8_M_BIASCM_P MOS'
xm10 net043 net050 VDDA VDDA sky130_fd_pr__pfet_01v8 l='MOSFET_10_1_L_gm2_P MOS' w='
MOSFET_10_1_W_gm2_P MOS*1' m='MOSFET_10_1_M_gm2_P MOS'
xm6 net050 VOUTN VDDA VDDA sky130_fd_pr__pfet_01v8 l='MOSFET_0_8_L_BIASCM_P MOS' w='
MOSFET_0_8_W_BIASCM_P MOS*1' m='MOSFET_0_8_M_BIASCM_P MOS'
xm5 VOUTN VOUTN VDDA VDDA sky130_fd_pr__pfet_01v8 l='MOSFET_0_8_L_BIASCM_P MOS' w='
MOSFET_0_8_W_BIASCM_P MOS*1' m='MOSFET_0_8_M_BIASCM_P MOS'
xm9 net063 VINP net31 net31 sky130_fd_pr__pfet_01v8 l='MOSFET_8_2_L_gm1_P MOS' w='
MOSFET_8_2_W_gm1_P MOS*1' m='MOSFET_8_2_M_gm1_P MOS'
xm8 DM_2 VINN net31 net31 sky130_fd_pr__pfet_01v8 l='MOSFET_8_2_L_gm1_P MOS' w='
MOSFET_8_2_W_gm1_P MOS*1' m='MOSFET_8_2_M_gm1_P MOS'
xm4 net31 net013 VDDA VDDA sky130_fd_pr__pfet_01v8 l='MOSFET_0_8_L_BIASCM_P MOS' w='
MOSFET_0_8_W_BIASCM_P MOS*1' m='4*MOSFET_0_8_M_BIASCM_P MOS'
xm3 VB3 net013 VDDA VDDA sky130_fd_pr__pfet_01v8 l='MOSFET_0_8_L_BIASCM_P MOS' w='
MOSFET_0_8_W_BIASCM_P MOS*1' m='MOSFET_0_8_M_BIASCM_P MOS'
xm2 DM_1 net013 VDDA VDDA sky130_fd_pr__pfet_01v8 l='MOSFET_0_8_L_BIASCM_P MOS' w='
MOSFET_0_8_W_BIASCM_P MOS*1' m='MOSFET_0_8_M_BIASCM_P MOS'
xm1 VB4 net013 VDDA VDDA sky130_fd_pr__pfet_01v8 l='MOSFET_0_8_L_BIASCM_P MOS' w='
MOSFET_0_8_W_BIASCM_P MOS*1' m='MOSFET_0_8_M_BIASCM_P MOS'
xm0 net013 net013 VDDA VDDA sky130_fd_pr__pfet_01v8 l='MOSFET_0_8_L_BIASCM_P MOS' w='
MOSFET_0_8_W_BIASCM_P MOS*1' m='MOSFET_0_8_M_BIASCM_P MOS'
```

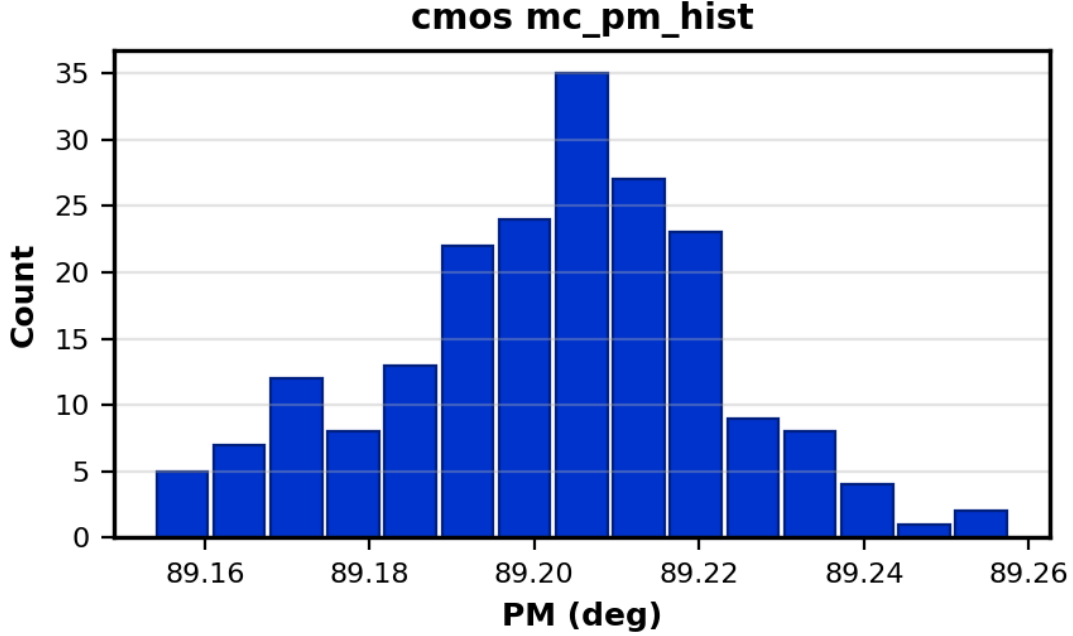


Figure 7: cmos LOT MC — PM histogram (F9), $N = 200$.

```

xm23 VOUT net049 GNDA GNDA sky130_fd_pr__nfet_01v8 l='MOSFET_23_1_L_gm3_NMOS' w='
MOSFET_23_1_W_gm3_NMOS*1' m='MOSFET_23_1_M_gm3_NMOS'
xm22 net049 net043 GNDA GNDA sky130_fd_pr__nfet_01v8 l='MOSFET_21_2_L_LOAD2_NMOS' w='
MOSFET_21_2_W_LOAD2_NMOS*1' m='MOSFET_21_2_M_LOAD2_NMOS'
xm21 net043 net043 GNDA GNDA sky130_fd_pr__nfet_01v8 l='MOSFET_21_2_L_LOAD2_NMOS' w='
MOSFET_21_2_W_LOAD2_NMOS*1' m='MOSFET_21_2_M_LOAD2_NMOS'
xm19 DM_2 VB4 GNDA GNDA sky130_fd_pr__nfet_01v8 l='MOSFET_17_7_L_BIASCM_NMOS' w='
MOSFET_17_7_W_BIASCM_NMOS*1' m='MOSFET_17_7_M_BIASCM_NMOS*8'
xm15 VOUTN VB3 DM_2 GNDA sky130_fd_pr__nfet_01v8 l='MOSFET_17_7_L_BIASCM_NMOS' w='
MOSFET_17_7_W_BIASCM_NMOS*1' m='MOSFET_17_7_M_BIASCM_NMOS*4'
xm20 net063 VB4 GNDA GNDA sky130_fd_pr__nfet_01v8 l='MOSFET_17_7_L_BIASCM_NMOS' w='
MOSFET_17_7_W_BIASCM_NMOS*1' m='MOSFET_17_7_M_BIASCM_NMOS*8'
xm16 net050 VB3 net063 GNDA sky130_fd_pr__nfet_01v8 l='MOSFET_17_7_L_BIASCM_NMOS' w='
MOSFET_17_7_W_BIASCM_NMOS*1' m='MOSFET_17_7_M_BIASCM_NMOS*4'
xm17 net54 VB4 GNDA GNDA sky130_fd_pr__nfet_01v8 l='MOSFET_17_7_L_BIASCM_NMOS' w='
MOSFET_17_7_W_BIASCM_NMOS*1' m='MOSFET_17_7_M_BIASCM_NMOS*4'
xm14 VB3 VB3 GNDA GNDA sky130_fd_pr__nfet_01v8 l='MOSFET_17_7_L_BIASCM_NMOS' w='
MOSFET_17_7_W_BIASCM_NMOS*1' m='MOSFET_17_7_M_BIASCM_NMOS'
xm12 VB4 VB3 net54 GNDA sky130_fd_pr__nfet_01v8 l='MOSFET_17_7_L_BIASCM_NMOS' w='
MOSFET_17_7_W_BIASCM_NMOS*1' m='MOSFET_17_7_M_BIASCM_NMOS*4'
xm18 net56 VB4 GNDA GNDA sky130_fd_pr__nfet_01v8 l='MOSFET_17_7_L_BIASCM_NMOS' w='
MOSFET_17_7_W_BIASCM_NMOS*1' m='MOSFET_17_7_M_BIASCM_NMOS*4'
xm13 DM_1 VB3 net56 GNDA sky130_fd_pr__nfet_01v8 l='MOSFET_17_7_L_BIASCM_NMOS' w='
MOSFET_17_7_W_BIASCM_NMOS*1' m='MOSFET_17_7_M_BIASCM_NMOS*4'
IO net013 GNDA 'CURRENT_0_BIAS'
CO net050 VOUT 'CAPACITOR_0'
.ends Fan_SMC_Pin_3

```

11.3 Fitted parameters (sky130)

```

* Fitted sky130 (scale=1.0u; W/L in um)
.param CURRENT_0_BIAS=50u
.param CAPACITOR_0=1p
.param CLOAD=10p
.param MOSFET_8_2_L_gm1_Pmos=0.5

```

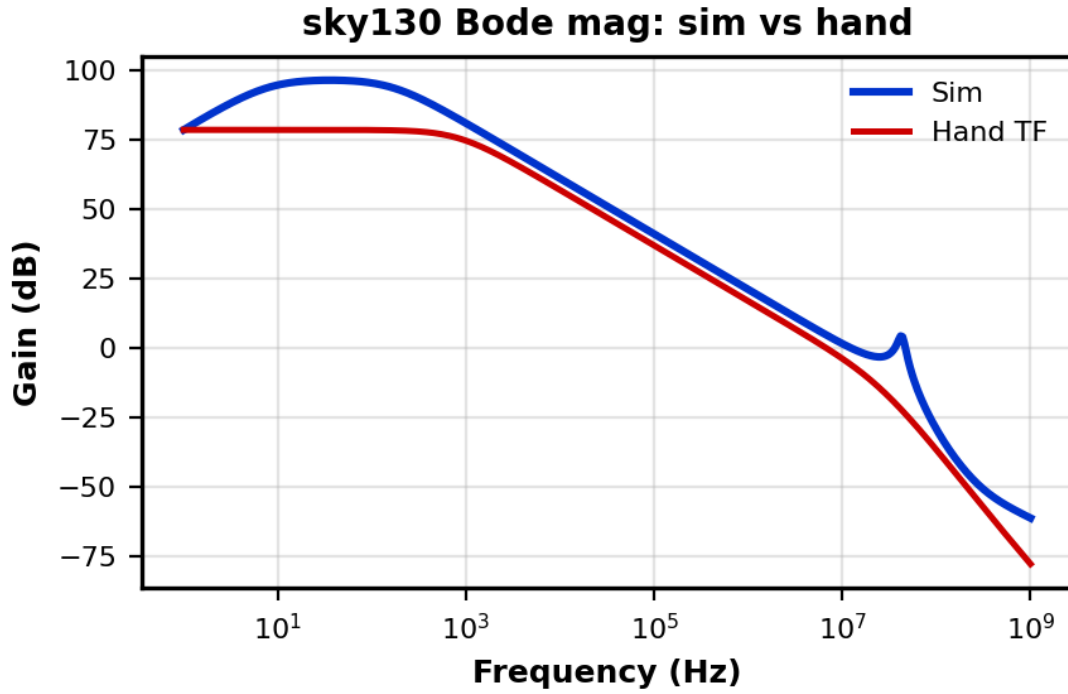


Figure 8: sky130 TT — Bode magnitude sim+hand (F1).

```
.param MOSFET_8_2_W_gm1_Pmos=20
.param MOSFET_8_2_M_gm1_Pmos=8
.param MOSFET_10_1_L_gm2_Pmos=0.5
.param MOSFET_10_1_W_gm2_Pmos=16
.param MOSFET_10_1_M_gm2_Pmos=8
.param MOSFET_11_1_L_gmf2_Pmos=0.5
.param MOSFET_11_1_W_gmf2_Pmos=10
.param MOSFET_11_1_M_gmf2_Pmos=16
.param MOSFET_23_1_L_gm3_Nmos=0.5
.param MOSFET_23_1_W_gm3_Nmos=20
.param MOSFET_23_1_M_gm3_Nmos=16
.param MOSFET_0_8_L_Biascm_Pmos=1
.param MOSFET_0_8_W_Biascm_Pmos=4
.param MOSFET_0_8_M_Biascm_Pmos=8
.param MOSFET_17_7_L_Biascm_Nmos=0.5
.param MOSFET_17_7_W_Biascm_Nmos=4
.param MOSFET_17_7_M_Biascm_Nmos=4
.param MOSFET_21_2_L_Load2_Nmos=0.5
.param MOSFET_21_2_W_Load2_Nmos=8
.param MOSFET_21_2_M_Load2_Nmos=4
```

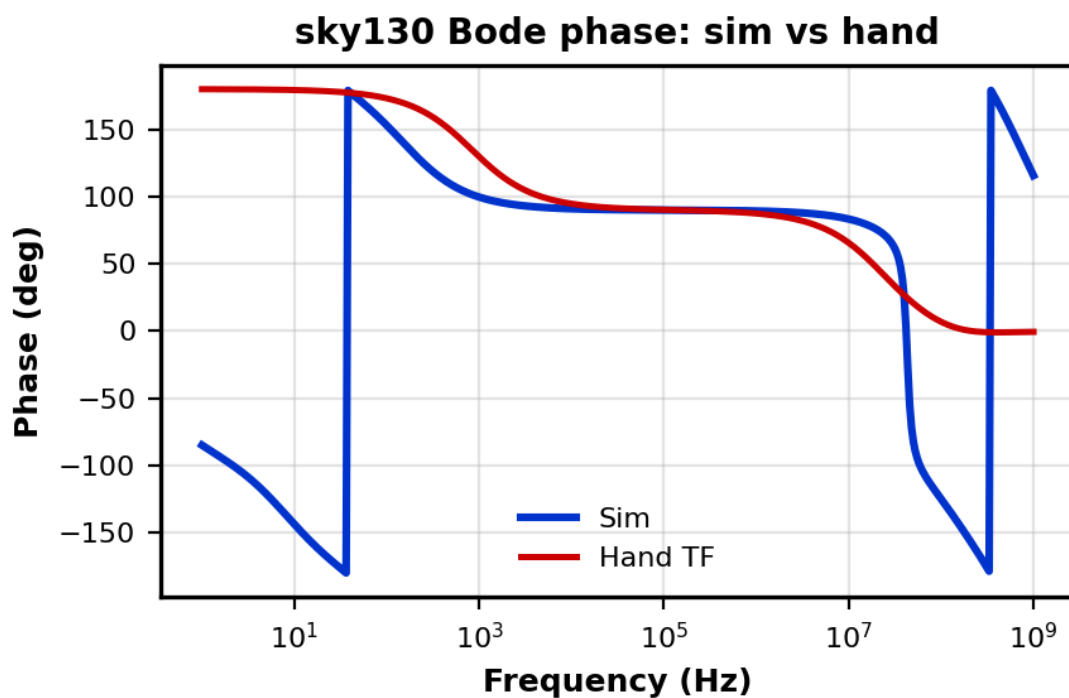


Figure 9: sky130 TT — Bode phase sim+hand (F2).

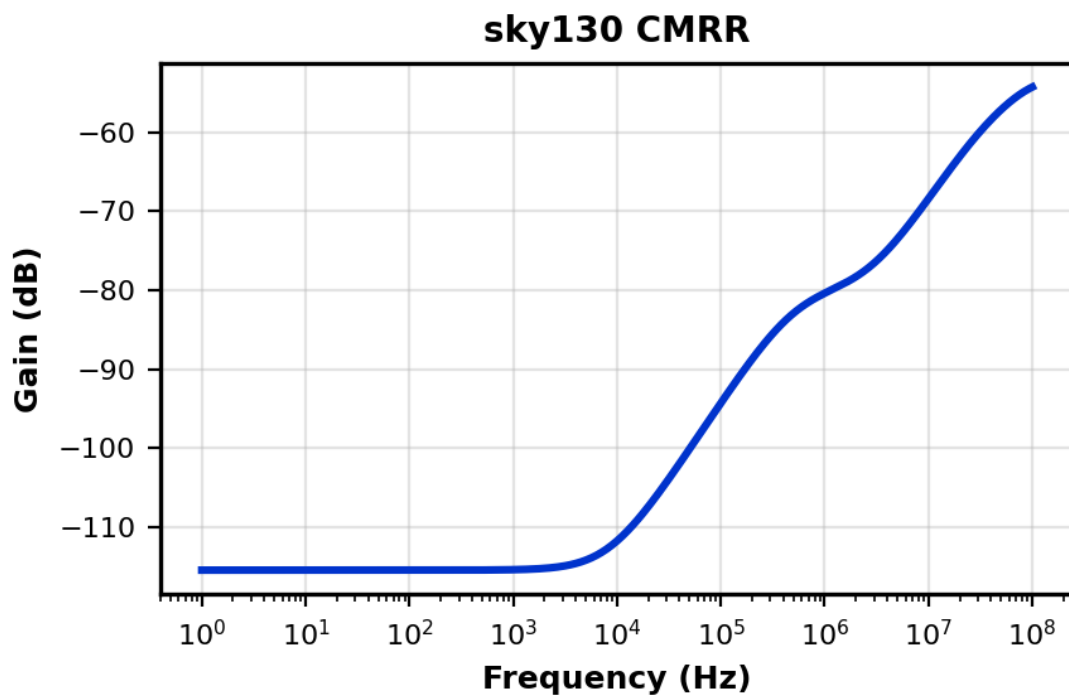


Figure 10: sky130 — CMRR (F3).

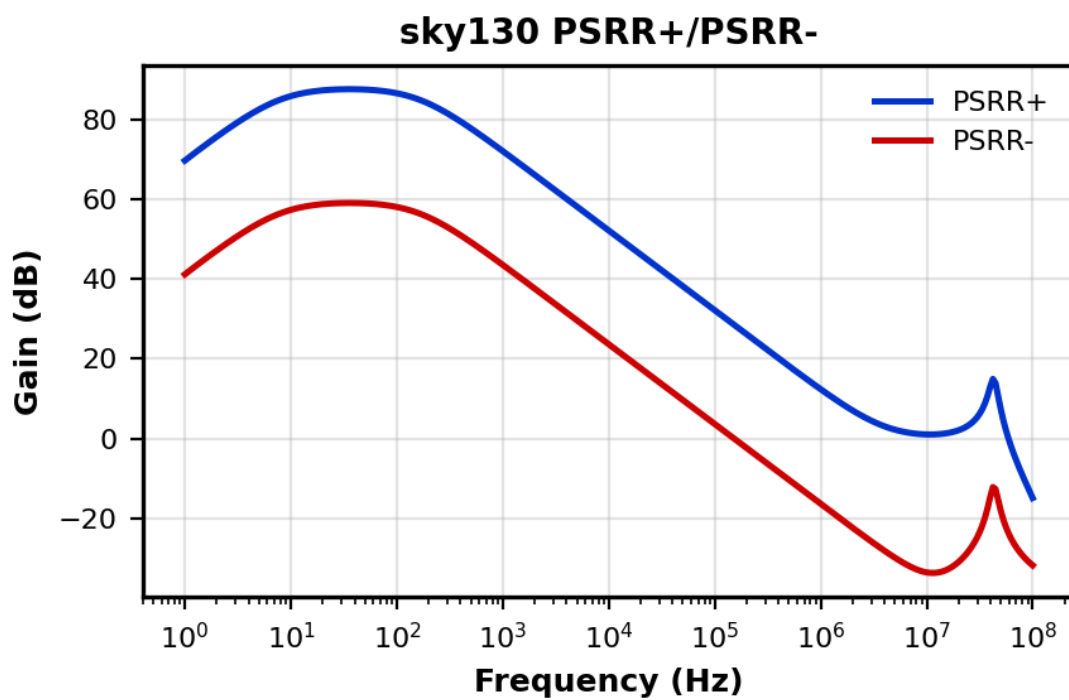


Figure 11: sky130 — PSRR (F4–F5).

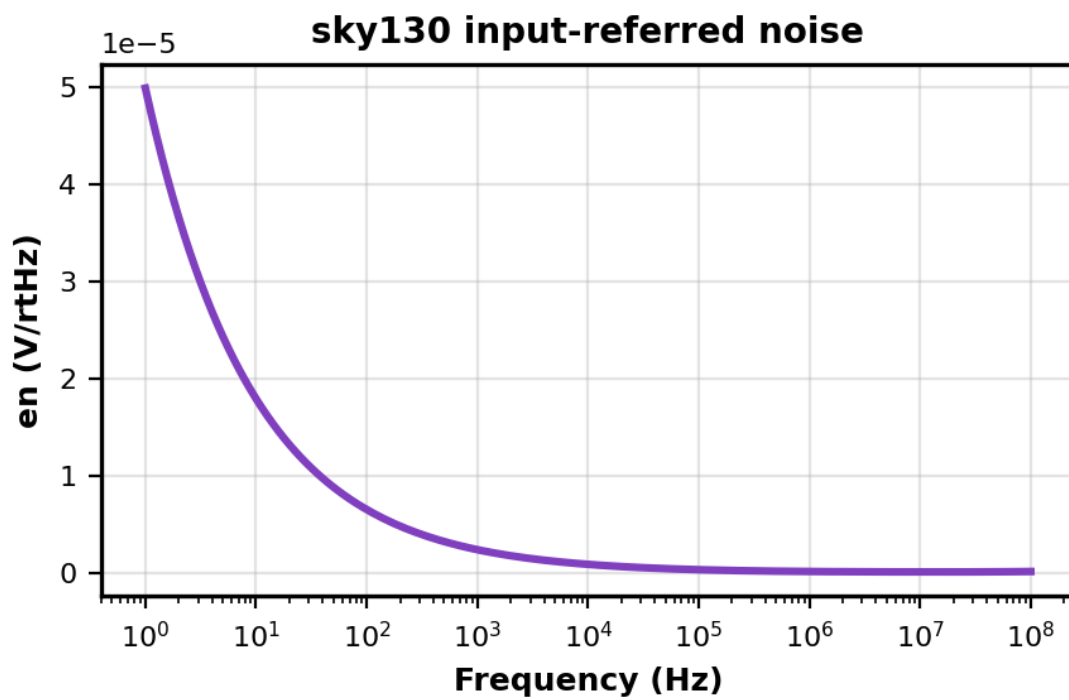


Figure 12: sky130 — noise density (F6).

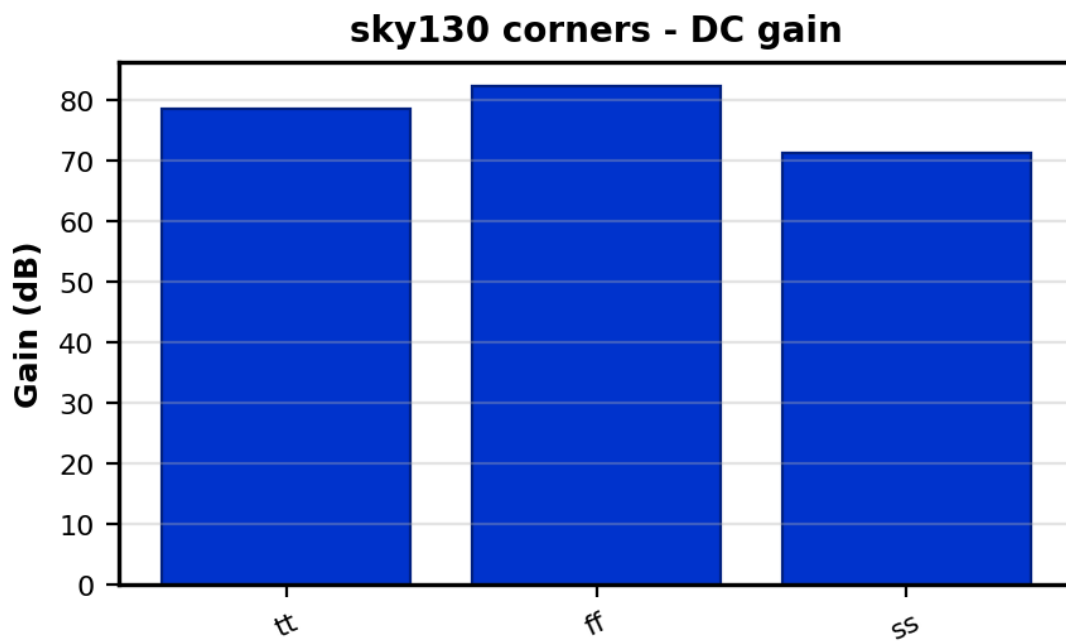


Figure 13: sky130 corners — DC gain (F7).

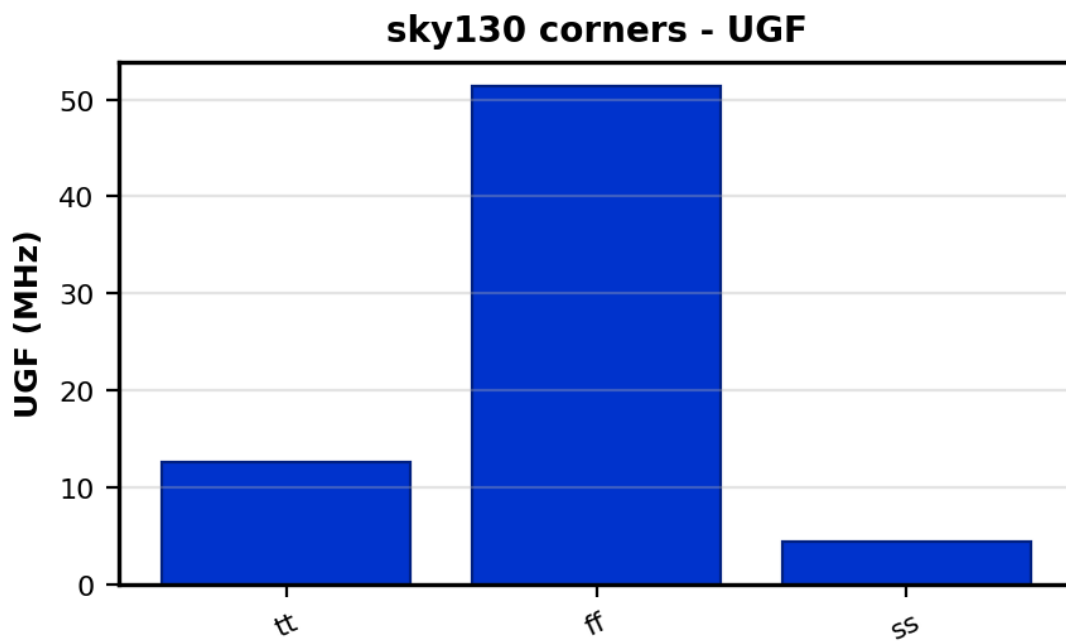


Figure 14: sky130 corners — UGF (F7).

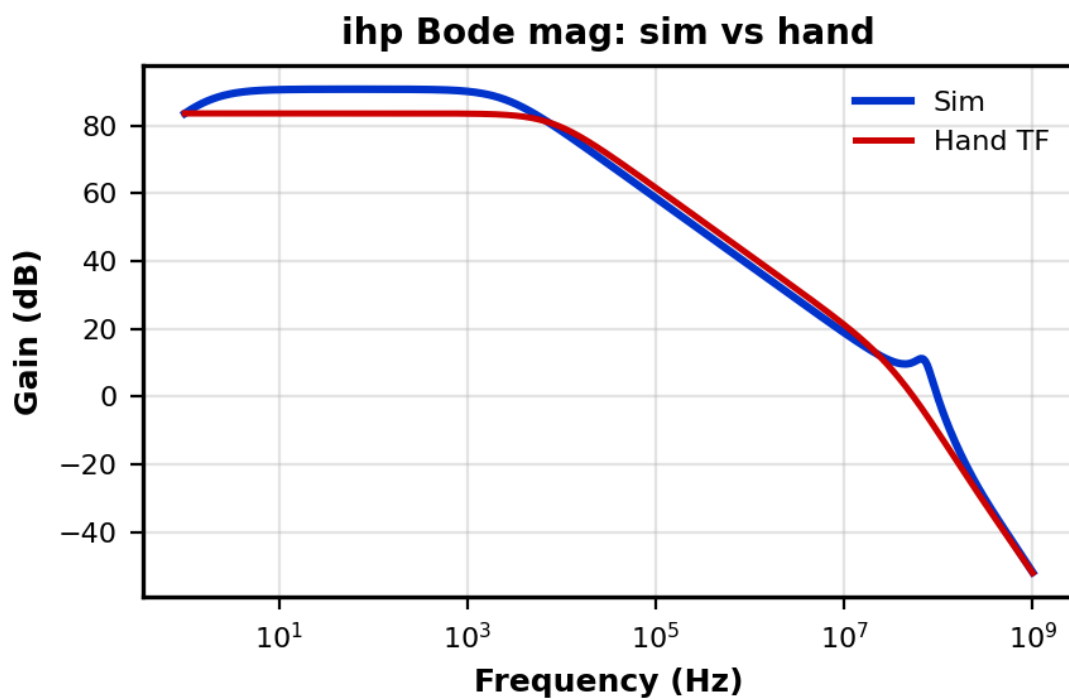


Figure 15: ihp — Bode magnitude sim+hand (F1).

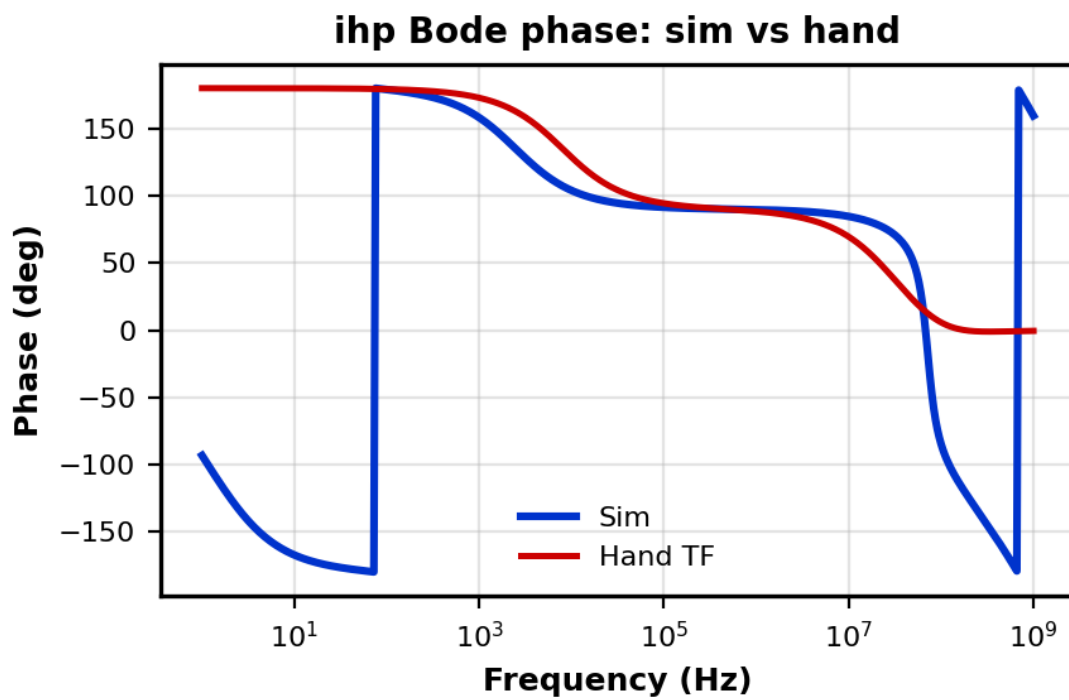


Figure 16: ihp — Bode phase sim+hand (F2).

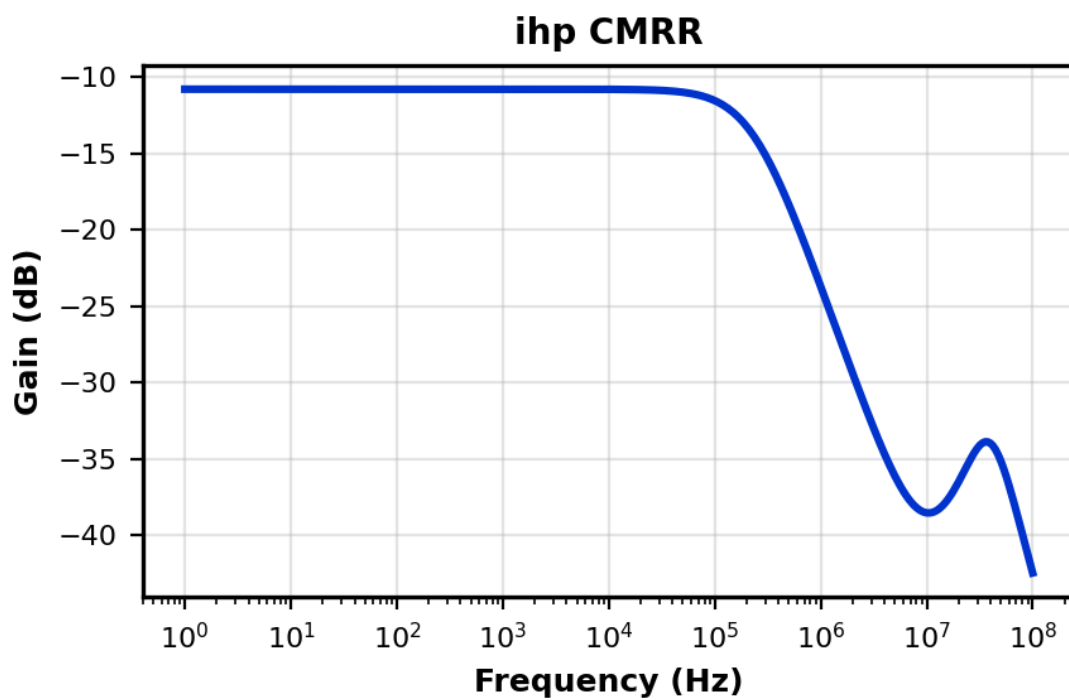


Figure 17: ihp — CMRR (F3).

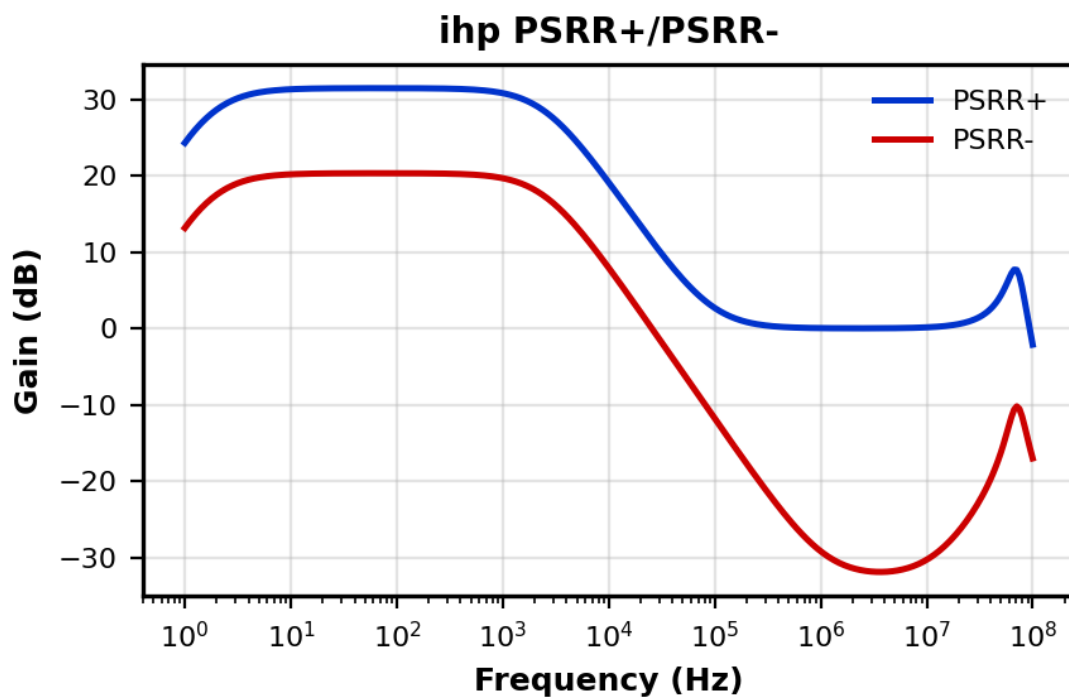


Figure 18: ihp — PSRR (F4–F5).

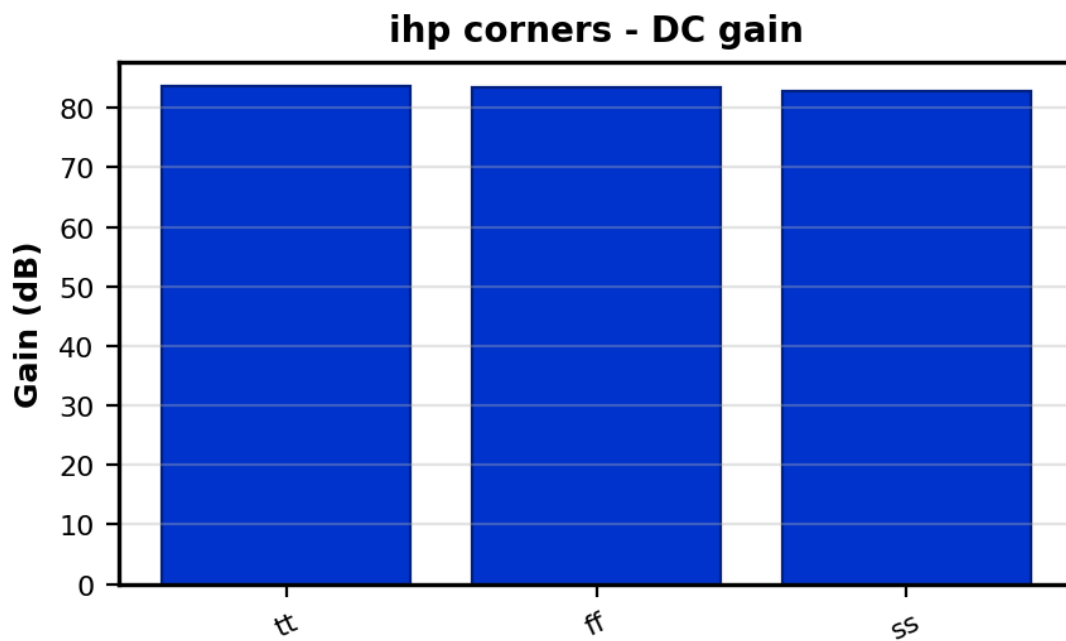


Figure 19: ihp corners — gain (F7).

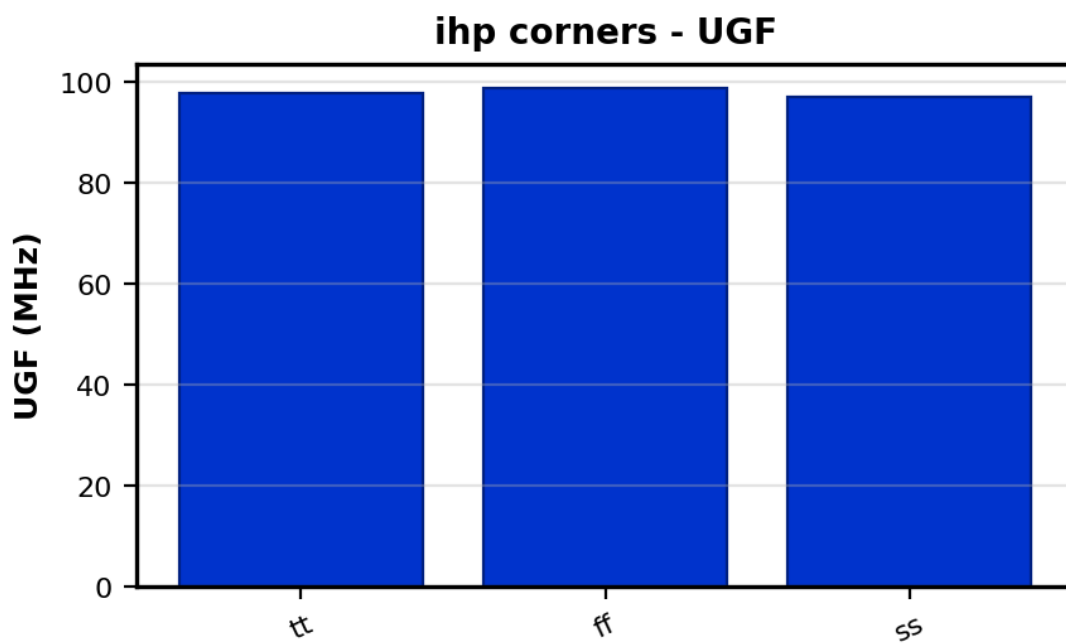


Figure 20: ihp corners — UGF (F7).

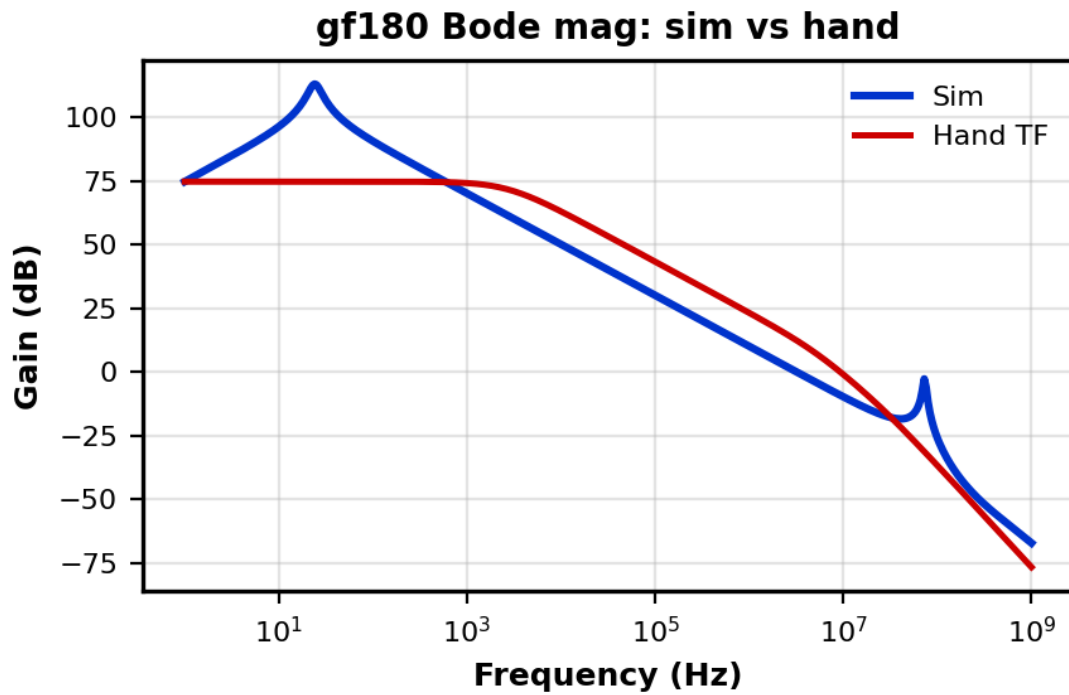


Figure 21: gf180 — Bode magnitude sim+hand (F1).

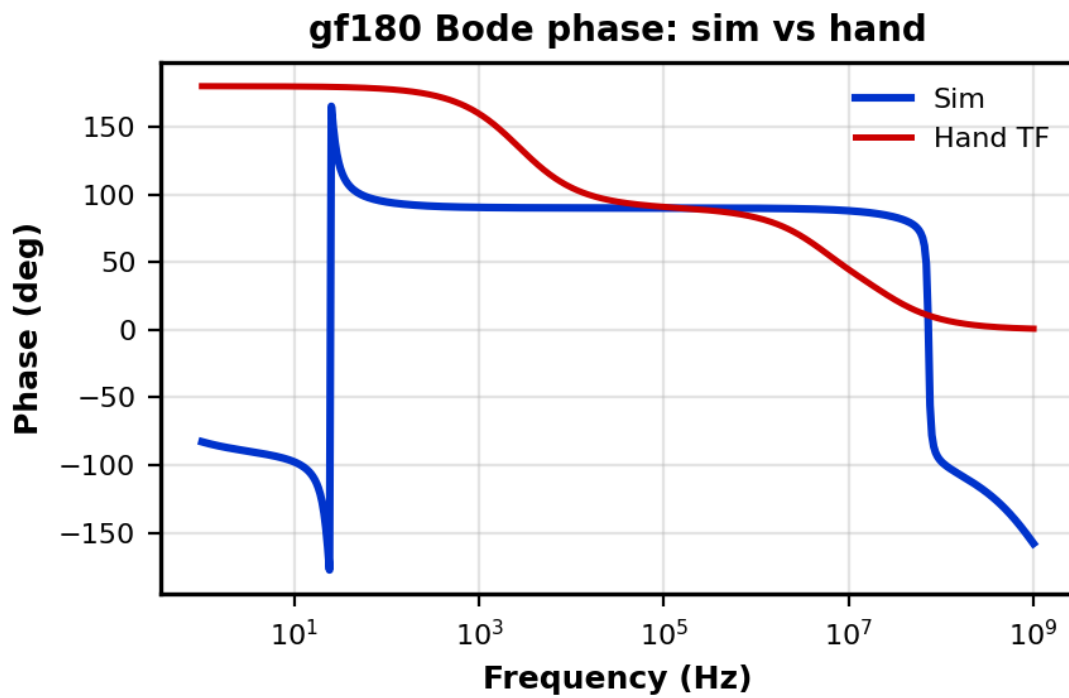


Figure 22: gf180 — Bode phase sim+hand (F2).

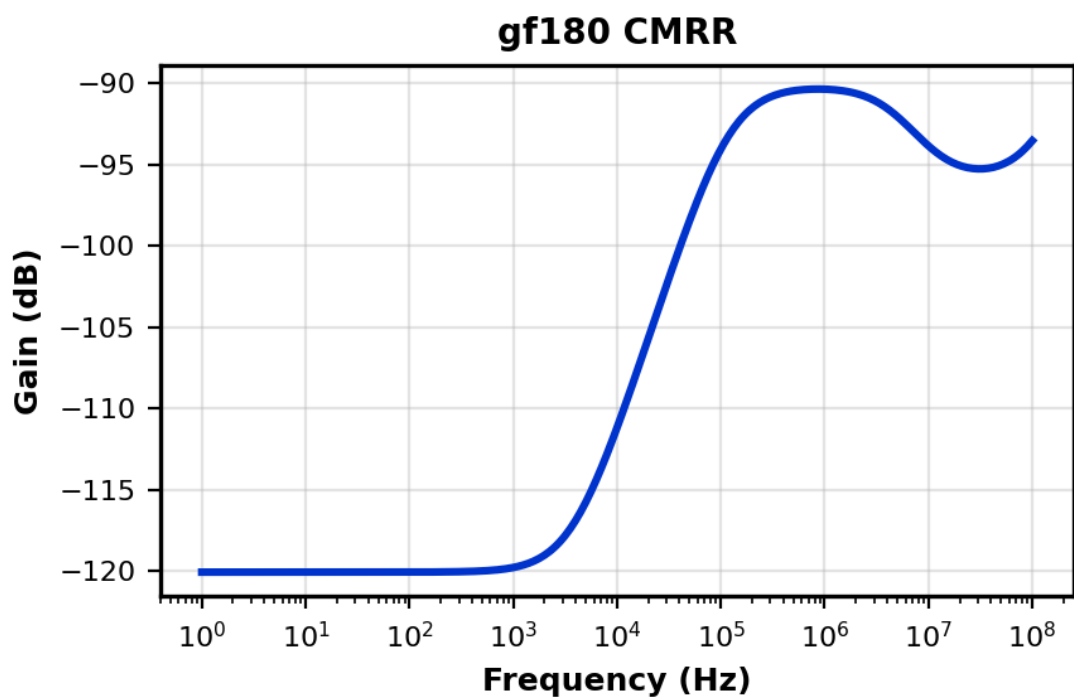


Figure 23: gf180 — CMRR (F3).

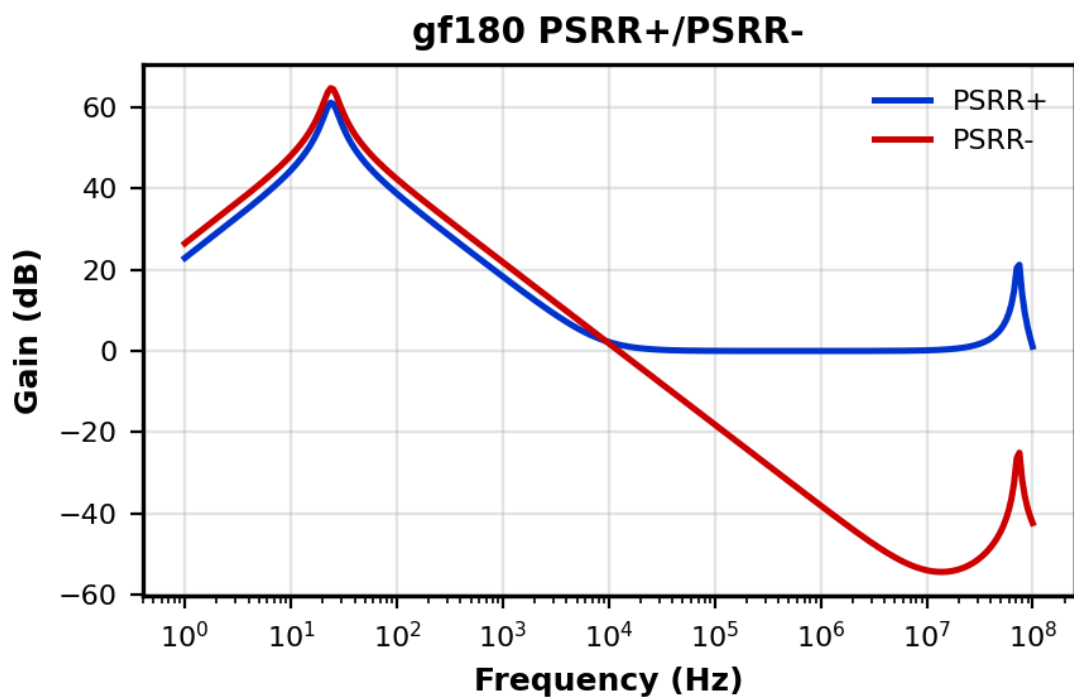


Figure 24: gf180 — PSRR (F4–F5).

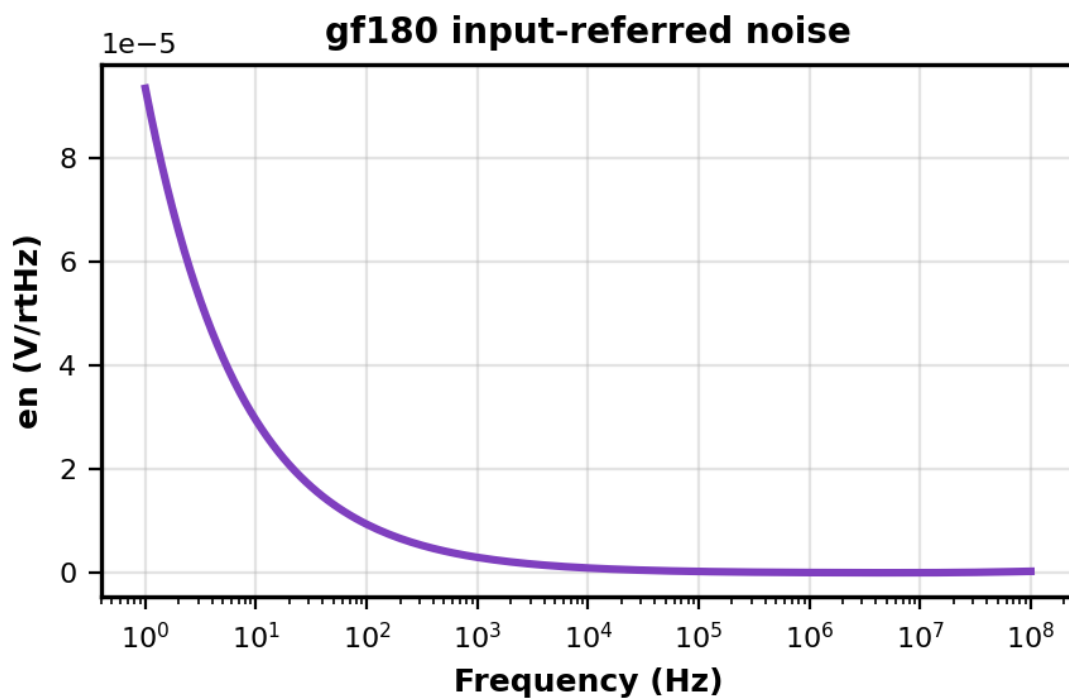


Figure 25: gf180 — noise (F6).

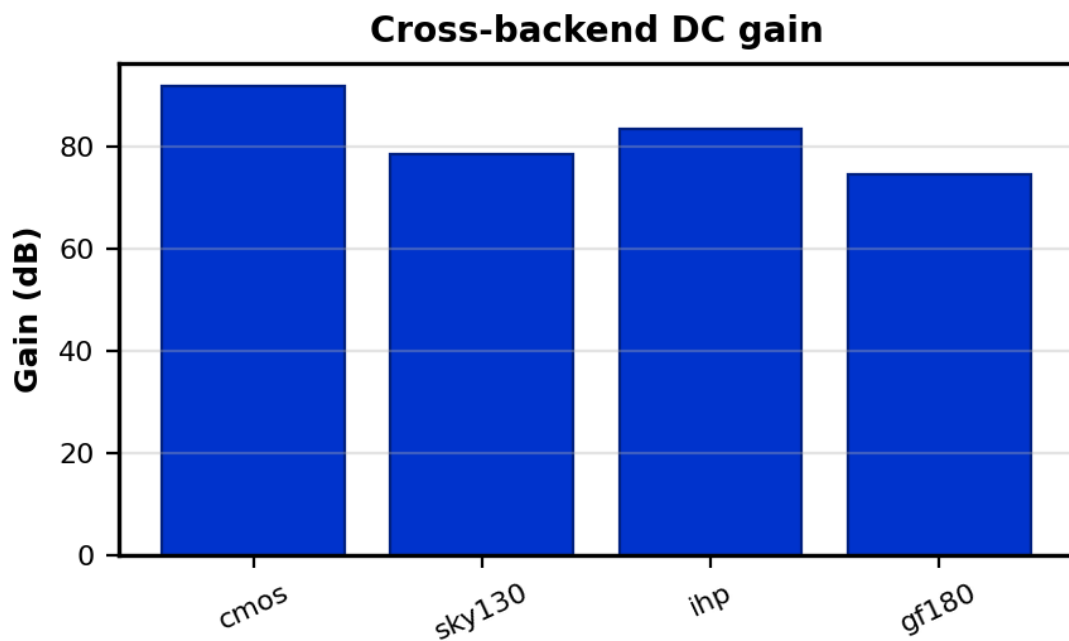


Figure 26: Cross-backend DC gain (F10).

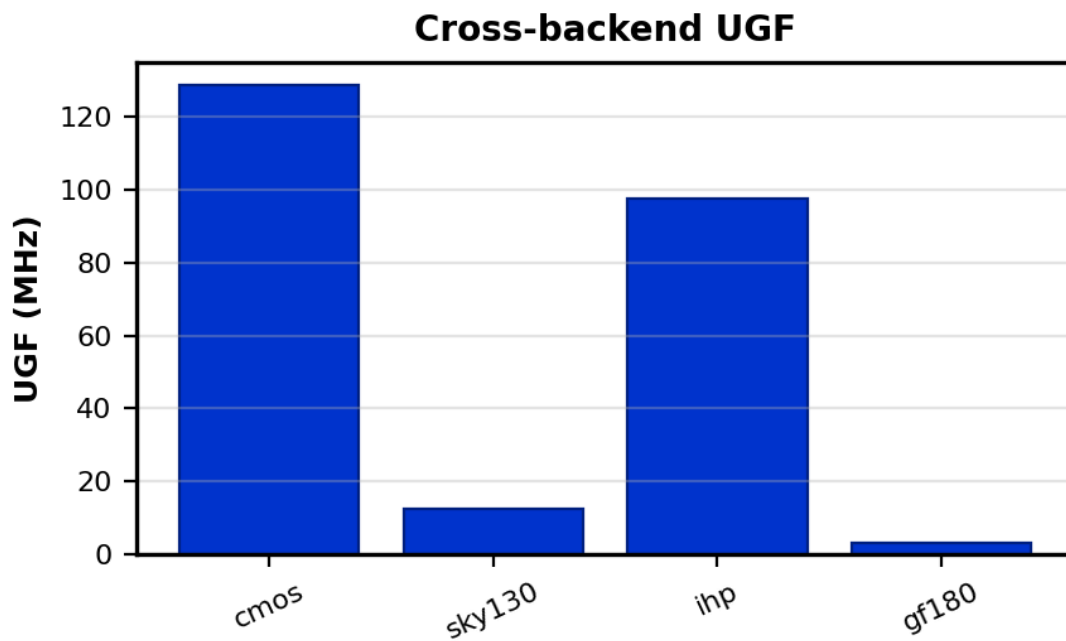


Figure 27: Cross-backend UGF (F10).

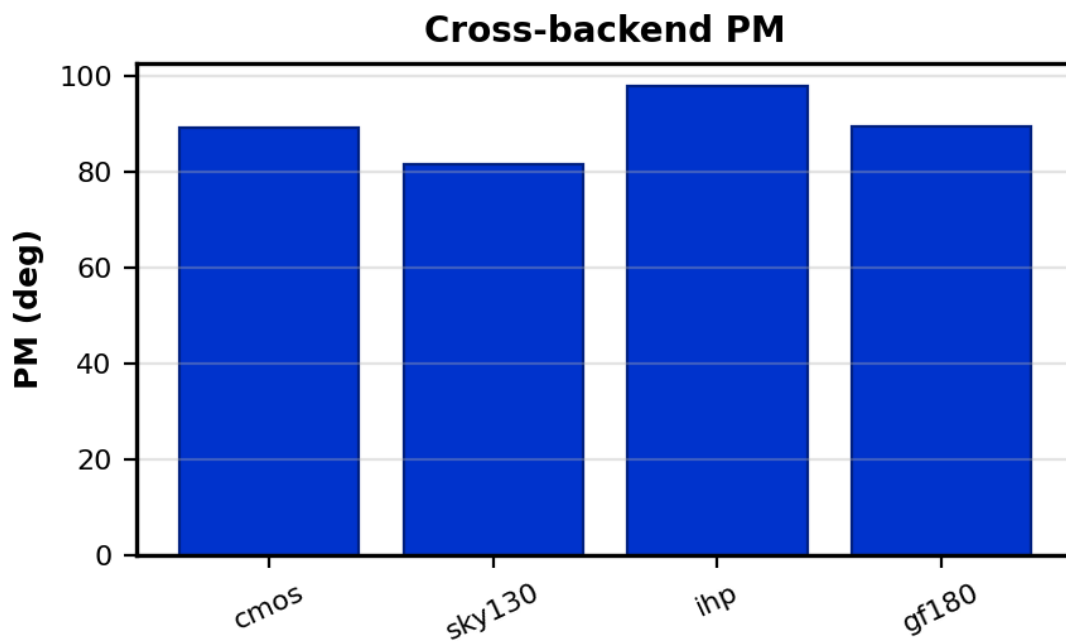


Figure 28: Cross-backend PM (F10).

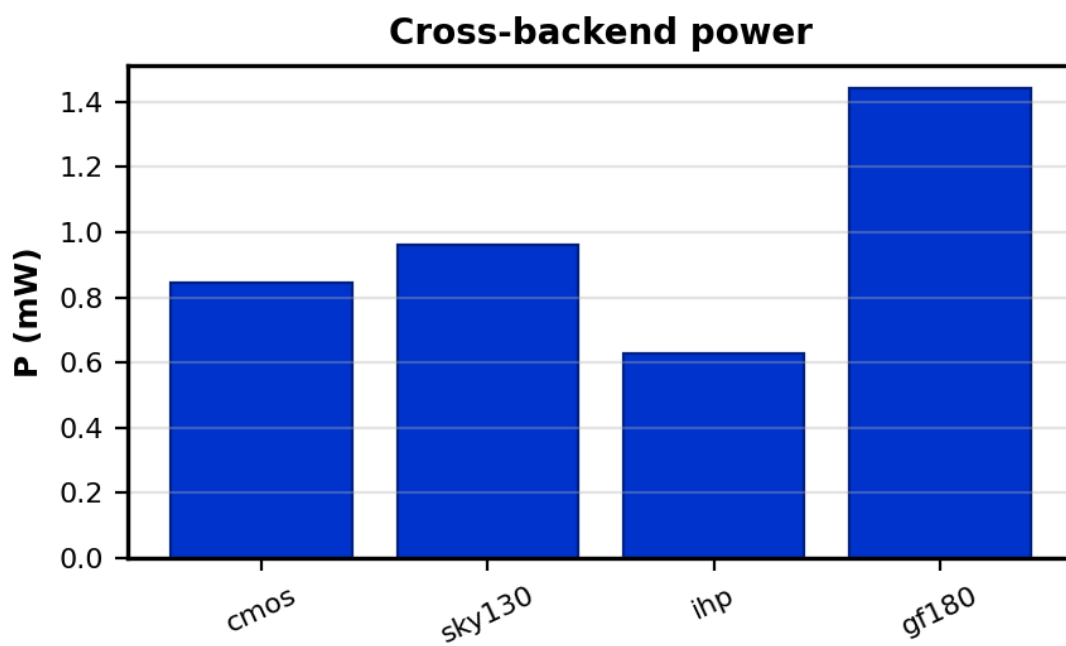


Figure 29: Cross-backend power (F10).